An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation

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Abstract

We propose a conditional scaling-spectrum framework, $H(q,t) = |q|^{n-1}/[s_0(t)e^{kq}]$. Prior work already establishes positive-q power-law generalized-Hurst spectra and nonlinear q-dependent multifractal moment functions; novelty is therefore not claimed for those ingredients. The proposed contribution is a signed-q three-coordinate decomposition—curvature n, baseline state s_0, and asymmetry k—together with prospective sequential moment-order validation before interpretation or extrapolation. Exploratory experiments include monofractal controls, IAAFT surrogate testing, controlled binomial and lognormal cascades, blind classification, parameter recovery, and held-out-q model comparison. The parameter n tracked independently controlled nonlinear scaling strength in two cascade families and survived a 500-surrogate transition test after Holm correction. The model is not universal: high-order p-model extrapolation was inferior to a quadratic competitor. Sequential validation achieved 88.5% blind applicability accuracy in a synthetic benchmark. The framework is presented as a testable family-conditional parameterization, not an established physical law.

Keywords: multifractal; generalized Hurst exponent; MF-DFA; intermittency; scaling spectrum; surrogate testing; moment-order prediction.

1 Background and prior work

Multifractal systems require moment-dependent scaling exponents rather than a single Hurst exponent. Multifractal detrended fluctuation analysis (MF-DFA), developed by Kantelhardt et al. [1], estimates generalized Hurst spectra in nonstationary time series and supports shuffled-signal comparisons that distinguish correlation-driven from distribution-driven multifractality.

This work originated from the elementary rational-power form $x^{n/s}$. Its ordinary antiderivative is standard calculus,

$$\int x^{n/s}\,dx = \frac{s}{n+s}x^{(n+s)/s} + C, \qquad n \neq -s, \tag{1}$$

and is not claimed as a new integration rule. The research question is whether separating a power-curvature coordinate from a response coordinate yields a useful, falsifiable generalized-moment spectrum.

Prior art constrains the novelty claim. Aligholi and Khandelwal [2] modeled fracture-surface multiscaling using $H(q) = Hq^{-\lambda_i}$, where λ_i is an intermittency exponent. For positive q, this is mathematically equivalent to the symmetric limit of the present model under $\lambda_i = 1-n$ and $H = 1/s_0$. Thus the positive-q power law and its basic intermittency interpretation are prior art. Grech [3] showed why whole-profile

information can matter beyond endpoint spectrum width. Recent work also targets reliability across moment order through threshold-sensitivity analysis [4] and consistency of generalized-Hurst estimation through focus regression [5].

2 Proposed scaling-spectrum family

For $q \neq 0$, define

$$\boxed{H(q,t) = \frac{|q|^{n-1}}{s_0(t)e^{kq}}} \tag{2}$$

and

$$\boxed{\zeta(q,t) = qH(q,t) = \frac{\mathrm{sgn}(q)|q|^n}{s_0(t)e^{kq}}}. \tag{3}$$

Taking logarithms gives

$$\boxed{\ln H(q,t) = (n-1)\ln|q| - \ln s_0(t) - kq}. \tag{4}$$

Equation (4) assigns distinct functional signatures to the parameters: n controls logarithmic curvature with moment magnitude, s_0 controls the baseline scaling level, and k controls signed-q asymmetry. Equal-magnitude positive and negative moments obey

$$\frac{H(-q,t)}{H(q,t)} = e^{2kq}. \tag{5}$$

For $n = 1$ and $k = 0$, $H(q,t) = 1/s_0(t)$ is independent of q, giving the monoscaling reference.

3 Prospective moment-order validation

A good in-sample fit is not sufficient. Let $\mathcal{Q}_m = \{q : 1 \leq |q| \leq m\}$ and let $\widehat{\boldsymbol{\theta}}_m = (\widehat{n}_m, \widehat{s}_{0,m}, \widehat{k}_m)$ be the fitted parameters. The framework predicts the next unseen shell,

$$\widehat{H}_m(\pm(m+1)) = \frac{(m+1)^{\widehat{n}_m - 1}}{\widehat{s}_{0,m}\exp[\pm\widehat{k}_m(m+1)]}. \tag{6}$$

Only after scoring Eq. (6) is the next shell revealed and the fit expanded. Interpretation is conditional on repeated prospective success over the q range of interest.

4 Experimental program

The exploratory program used MF-DFA-derived spectra together with synthetic controls and known cascade generators. The white-noise, $1/f$, Cauchy, mono-to-multifractal transition, and healthy RR reference series were obtained from the PhysioNet Multiscale Multifractal Analysis (MMA) v1.0.0 resource [6, 7, 8]; the resource is distributed under the Open Data Commons Attribution License v1.0. Tests included Gaussian/Brownian and $1/f$ controls; a healthy RR-interval series; a heavy-tailed Cauchy reference; a mono-to-multifractal transition signal; binomial p-model cascades; conservative lognormal cascades; IAAFT surrogates; blind mixed-family classification; direct parameter-recovery simulations; and held-out-q comparison with competing three-parameter spectrum models. A frozen end-to-end reproduction pipeline was subsequently executed from source data and fixed generators. The

headline cascade, surrogate, recovery, benchmark, blind-classification, and sequential-validation statistics reproduced. Reference-signal point estimates changed slightly after standardizing the final pipeline to 10 MF-DFA scales; Table 1 reports the frozen-pipeline values.

5 Results

5.1 Reference behavior

Table 1: Frozen-pipeline fits to reference and real-system spectra.

Dataset	n	s_0	k	RMSE
White noise	0.9979	2.0547	0.00324	0.00052
$1/f$ noise	0.9958	1.0104	0.00929	0.00091
Healthy RR series	0.9737	1.1701	0.03100	0.00542
Cauchy multifractal	0.6529	1.1679	0.15096	0.03273

Near-monofractal controls approach $n = 1$ and $k = 0$, whereas the heavy-tailed multifractal reference moves far from that point.

5.2 IAAFT transition confirmation

Table 2: 500-surrogate IAAFT confirmation on the known transition signal.

Window	Metric	Observed	IAAFT 95% interval	Raw p	Holm p
2048	Δn	-0.1032	$[+0.0360, +0.1293]$	0.001996	0.00798
2048	Δk	+0.0478	$[+0.0464, +0.1090]$	0.0539	0.0998
4096	Δn	-0.0327	$[+0.0541, +0.1398]$	0.001996	0.00798
4096	Δk	+0.0351	$[+0.0354, +0.0950]$	0.0499	0.0998

With 500 IAAFT surrogate pairs, Δn survived Holm correction at both preselected larger windows, whereas Δk did not.

5.3 Controlled cascade-strength sweeps

Across 850 p-model realizations spanning $p = 0.49$ to 0.10, n correlated strongly and negatively with the analytically known singularity-spectrum width

$$\Delta\alpha = \left|\log_2 \frac{1-p}{p}\right|, \tag{7}$$

with Spearman $\rho = -0.9902$ and Pearson $r = -0.9701$. A descriptive quadratic calibration explained $R^2 = 0.9862$ of between-level mean-n variation.

Across 700 independent conservative lognormal cascades, n correlated with known generator strength $2\sigma^2$ at Spearman $\rho = -0.8725$ and Pearson $r = -0.9861$, with descriptive calibration $R^2 = 0.9892$. The numerical mappings differ between families; n is therefore not a universal conversion of multifractality into one scalar.

5.4 Blind discrimination and complementarity

A 400-signal blind mixed-family experiment gave 58.0% exact four-class accuracy using n alone versus a 25% random baseline; 92.7% of cases were within one adjacent strength class, and Spearman correlation between hidden strength order and n was -0.741. However, an established spread proxy $\Delta H = H(-5) - H(5)$ was more predictive by itself. Five-fold ordinal RMSE was 0.789 for n alone, 0.459 for ΔH alone, 0.455 for $\Delta H + n$, and 0.399 for ΔH plus a nonlinear residual plus n.

5.5 Parameter identifiability

Table 3: Recovery correlations from 30,000 directly generated spectra.

Log-noise SD	r_n	r_{s_0}	r_k
1%	0.9990	0.9998	0.9998
3%	0.9910	0.9978	0.9985
7%	0.9544	0.9888	0.9916

The transformed regression in Eq. (4) uses the basis functions $\ln|q|$, 1, and q. Over the tested moment set the design matrix had condition number approximately 8.43.

5.6 Competing-model benchmark and adverse result

Table 4: Held-out moment-order extrapolation against competing three-parameter spectra.

Family	Model	Held-out RMSE	Win rate
Lognormal	n/s	0.0213	59.5%
Lognormal	quadratic log-H	0.0291	21.5%
p-model	n/s	0.5880	21.5%
p-model	quadratic log-H	0.2089	75.5%

The adverse p-model result is central: the proposed family is not universal. It extrapolated well on the tested lognormal family but was materially inferior to a quadratic log-spectrum at high p-model moment orders.

5.7 Applicability testing

Training AIC and local n-stability were weak applicability gates, at approximately 58% and 60% blind accuracy. The prospective sequential rule was substantially stronger: 88.5% accuracy, 86.1% balanced accuracy, 85.5% precision when accepted, 97.5% sensitivity, and 74.7% specificity. The simulation-specific early-shell maximum log-error threshold was 0.04885; this numerical threshold is not proposed as universal.

6 Mathematical interpretation

For $k = 0$ and $q > 0$,

$$\zeta(q) = \frac{q^n}{s_0}, \qquad \zeta''(q) = \frac{n(n-1)}{s_0} q^{n-2}. \tag{8}$$

Hence $0 < n < 1$ implies $\zeta''(q) < 0$, i.e. a concave positive-q spectrum. With $k \neq 0$, curvature contains additional exponential terms, so n alone is not a complete multifractality measure. In the positive-q symmetric limit, $1 - n$ is equivalent to the intermittency exponent λ_i of Ref. [2].

7 Discussion and domain of validity

The evidence supports a limited interpretation. The parameter n is identifiable, repeatedly approaches unity in monoscaling controls, and responds systematically to independently controlled nonlinear scaling strength in two different cascade families. Yet n alone is weaker than ΔH as a generic strength measure, and the p-model demonstrates that a locally good fit can belong to the wrong global geometry. The framework should therefore be used conditionally: if sequential moment-order prediction succeeds, n can be interpreted as a compact power-curvature coordinate over the validated q range.

This predictive gate is the most important methodological result. It makes the framework falsifiable from its own moment-order behavior rather than allowing any well-fitted spectrum to be declared a member of the family.

8 Novelty position

The literature search found direct precedent for $H(q) \propto q^{-\lambda}$, q-power multifractal moment functions, whole-profile multifractality measures, and q-dependent reliability methods. No exact precedent was found in the searched literature for the complete form in Eq. (2) combined with sequential held-out-q prediction as a family-membership or falsification gate. This supports a claim of potential framework novelty, not proof of exhaustive scholarly or patent novelty.

A defensible contribution statement is: building on power-law generalized-Hurst representations, the framework introduces a signed-moment three-coordinate spectrum and requires prospective moment-order prediction before parameter interpretation or extrapolation.

9 Data availability and attribution

The five reference time series used in the frozen validation pipeline—monofractal white noise, $1/f$ noise, multifractal Cauchy-distributed noise, the mono-to-multifractal phase-transition signal, and the healthy RR-interval series—were obtained from the PhysioNet *Multiscale Multifractal Analysis* v1.0.0 resource [6, 7, 8]. PhysioNet identifies the resource as open access under the Open Data Commons Attribution License v1.0 and requests citation of the original MMA publication [6]. Copies of the specific reference inputs used here are included in the accompanying Zenodo research package with source attribution preserved.

Exploratory financial tests used the public `spy_data.csv` file from the OStochastic *Daily-SPY-data-from-2000-2025* GitHub repository [9]. The repository describes the file as daily SPDR S&P 500 ETF (SPY) market data from 2000 onward. Those financial tests are supporting exploratory analyses rather than the principal frozen reference-signal validation. Because the GitHub repository page located for this revision does not state a file license, inclusion of that third-party CSV in redistributed archives should be treated separately from the PhysioNet data; users should consult the upstream repository for current reuse terms.

10 Limitations and reproducibility

The principal experiments were consolidated into a frozen end-to-end Python pipeline and rerun before this Version 1.0 deposit. The rerun reproduced the manuscript headline values for the 500-IAAFT confirmation, p-model and lognormal strength correlations, blind classification/complementarity tests, 30,000-trial parameter recovery, held-out-q benchmark, and sequential applicability test. The final frozen MF-DFA implementation uses 10 scales, which produced the reference-signal values in Table 1. Further publication-quality work should still use larger surrogate convergence studies, quantify scale-selection uncertainty, reproduce real-data results on independent datasets, and compare against additional established multifractal models. The simulation-specific sequential-error threshold must be recalibrated for other settings. The frozen code, source data available for redistribution in the research package, and generated result tables are archived with this preprint at Zenodo DOI 10.5281/zenodo.23111783. The present results establish a reproducible research hypothesis and evidence base, not a confirmed physical law.

11 Conclusion

The n/s scaling-spectrum framework is a candidate family-conditional model of generalized moment scaling. Its positive-q power-law core has prior precedent, while the proposed signed-q decomposition and prospective moment-order validity test appear distinct in the literature searched. Controlled experiments support n as an identifiable curvature coordinate that tracks nonlinear scaling strength within compatible families. Equally important, adverse p-model results show where the geometry fails. The framework's scientific value therefore lies not in universality, but in a compact representation coupled to an explicit mechanism for rejecting its own applicability.

References

[1] J. W. Kantelhardt, S. A. Zschiegner, E. Koscielny-Bunde, S. Havlin, A. Bunde, and H. E. Stanley, "Multifractal detrended fluctuation analysis of nonstationary time series," *Physica A*, vol. 316, pp. 87–114, 2002. doi:10.1016/S0378-4371(02)01383-3.

[2] S. Aligholi and M. Khandelwal, "Intermittency of Rock Fractured Surfaces: A Power Law," *Water*, vol. 14, no. 22, 3662, 2022. doi:10.3390/w14223662.

[3] D. Grech, "Alternative measure of multifractal content and its application in finance," *Chaos, Solitons & Fractals*, vol. 88, pp. 183–195, 2016. doi:10.1016/j.chaos.2016.02.017.

[4] M. Mangalam, E. Watanabe, and K. Kiyono, "Additomultiplicative Cascades Govern Multifractal Scaling Reliability Across Cardiac, Financial, and Climate Systems," *Entropy*, vol. 28, no. 3, 359, 2026. doi:10.3390/e28030359.

[5] W. Schadner, "Focus regression for multifractal analysis," *Chaos, Solitons & Fractals*, vol. 209, 118368, 2026. doi:10.1016/j.chaos.2026.118368.

[6] J. Gierałtowski, J. J. Zebrowski, and R. Baranowski, "Multiscale multifractal analysis of heart rate variability recordings with a large number of occurrences of arrhythmia," *Physical Review E*, vol. 85, 021915, 2012. doi:10.1103/PhysRevE.85.021915.

[7] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[8] T. Pollard et al., "PhysioNet as a global platform for biomedical research," *Nature Health*, 2026. doi:10.1038/s44360-026-00096-z.

[9] OStochastic, "Daily-SPY-data-from-2000-2025," GitHub repository, accessed October 2026. https://github.com/OStochastic/Daily-SPY-data-from-2000-2025.